UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 1: Programming Fundamentals (CS-308)

Basic Information			
Lab #:	1	**Teacher:**	Arsal Mahmood
Objectives	1. Basic programming logics 2. Writing simple programs 3. Variables, datatypes and their uses		
Registration#		**Student Name**	
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : Variables and Datatypes

1. Write a simple program to print your name on screen.
2. Write a program to input a number from the user, double it and display it on the screen.
3. Write a program to input two numbers and display their sum.
4. Write a program to calculate the area of a rectangle by taking length and width as input from the user.
5. Get user given values in variables *a,b* and *c* and display answer for the following equation:

$$8ax^2 + 6bx + 4c$$

6. Write a program to calculate the average of three numbers entered by the user.
7. Write a program to input the radius of a circle and calculate its area.
8. Write a program to convert temperature from Celsius to Fahrenheit.
9. Input username, age, gender and city and store them in relevant variables. Then display on screen like: Hello **"name",** You're a **"age"** years old **"gender"** living in **"city"**.
10. Develop a calculator which can handle two user given numbers. The calculator can perform addition, subtraction, multiplication and division and display all the results on screen.
11. Write a program that takes principal amount, rate of interest, and time from the user and calculates the simple interest using the formula:

$$SI = \frac{P \times R \times T}{100}$$

The program should display the calculated simple interest.

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12. Write a program that asks the user to enter the length of one side of a square. The program should then calculate: Display both the area and perimeter.

Perimeter = 4 × side

Area = side × side

13. Write a program that takes two numbers as input from the user. The program should calculate the square of each number and then add the two squares together. Display the final result. Example formula:

$$Result = a^2 + b^2$$

14. Write a program that asks the user to enter the price of an item. The program should calculate the total bill amount after adding 10% tax. Display the final bill amount including tax.

$$Total = Price + (Price \times 0.10)$$

15. Write a program that takes the base and height of a triangle as input from the user. The program should calculate the area of the triangle using the formula. Display the calculated area of the triangle.

$$Area = \frac{1}{2} \times base \times height$$

{end of Lab}